

3.7 Vectors

Pure Mathematics 3 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A line in 3D has $r=a+\lambda b$. Dot product $a \cdot b = |a||b|\cos\theta$ gives angle and perpendicularity. To intersect two lines, solve all three component equations for their separate parameters; a pairwise apparent intersection may be skew if the third fails.

Worked example: follow the reasoning

$(1,2,3) \cdot (2,-1,0) = 0$, so direction vectors are perpendicular.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 For $a=(1,2,2)$ and $b=(2,0,0)$, find $\cos\theta$ between them.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$a \cdot b = 2$ [1], $\text{abs}(a) = 3$ and $\text{abs}(b) = 2$ [1]; $\cos\theta = 2/6 = 1/3$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.